

Station 3: Camera obscura

Focus the 8×10 under a cloth and photograph the glass. Then into the dark room with a partner: one holds the acrylic, one shoots. Then pick a digital chain: the infinite loop, the hand-held lens, or the projected ghost.

Where your shots go: The 8×10 ground-glass photo. The dark-room acrylic photo. At least one from the digital chain you picked. Phone shots to the class image channel; projector and camera shots go to the wall when you press **Capture** on the station computer.

Terms to master

Term	In one line
Camera obscura	A dark room or box with a small hole (or lens) in one wall, on the opposite wall of which the outside scene appears upside down.
Ground glass	A sheet of glass frosted on one side so that a lens's image forms on it and can be seen from behind; the viewfinder of a large-format camera.
Image plane	The surface behind the lens where the image comes to focus: the ground glass, the film, the sensor. Whatever sits there is what records.
Inversion	The image a lens forms is upside down and reversed left to right. Every camera flips it back for you; the ground glass doesn't.
Vignetting	The image getting darker toward the edges and corners, because the lens throws less light off-axis or something physically blocks it.
Hot spot	The bright center of a rear-projected image, where the lens's light comes straight through the screen at your eye.
Large format	Cameras and lenses made for sheets of film 4×5 inches and up; huge image circles, long focal lengths, and very shallow focus.
Projector	A lens that throws a picture out onto a wall or a person instead of taking one in: the camera obscura run backwards. The magic lantern's descendant.
Video feedback	Pointing a camera at the screen showing its own live picture, so the image contains itself, repeating inward.
Lens whacking	Holding a lens off the camera body, tilted and moving, so light leaks in and the plane of focus tilts. A deliberate misuse.

Gear: the 8×10 large-format camera on its stand with its ground glass and a velvet cloth; the dark room, with lenses on stands and sheets of frosted acrylic; a camera and a screen showing its own feed; a lens that comes off its body; the projector and the green-screen stage; your phone; the prints on the table (Frisius 1545, Kircher's 1646 walk-in cube, Zahn's 1685 reflex box).

What it's about

The first experiment is the oldest one: the world projected onto a wall, captured in a drawing, augmented in a painting. Every stage adds something, and the picture at the end is made of all of them. Take that as the inspiration for a broad array of related experiments: chains, loops, composites, artifacts. The camera obscura is where you start; where the chain goes after that is up to you.

Recipe

1. **The 8×10.** Don't climb into the big dark box; just put the velvet cloth over your head and the back of the camera. The image on the ground glass is upside down and reversed; that is correct. Rack the focus until your subject's eyes are sharp. Then take a picture of the ground glass with your phone.
2. **The dark room, with a partner.** Go in together. One of you holds the frosted acrylic behind the lens (and the lens too, if it needs holding) and slides it until the image snaps sharp. The other photographs the acrylic with a phone. Three people if it takes three. Swap roles. Notice the hot spot in the middle, the dark edges, the grain of the sheet; that is the sheet adding its stage, and the phone adds another when it exposes and white-balances.
3. **Hop out and pick one digital chain** (or more, if there's time):
 - **The infinite loop.** Point a camera at the screen that's showing its own feed. Tilt it a few degrees; zoom in slowly. Then stand in front of that camera, pose, and move; watch yourself repeat into the tunnel. Photograph or film the screen.
 - **The hand-held lens.** With a staff member, take the lens off a camera body and hold it by hand just in front of the open mount. Tilt it, move it in and out. You are after the artifacts you got in the dark box, light leaks and odd planes of focus, made with a modern camera. Capture the ones you'd keep.
 - **The ghost.** Take the projector while a partner gets on the green-screen stage. The projector throws their image as a ghostly apparition; put it on the wall, on a person, on the acrylic, and work it into your experiment. The station camera's feed is on the computer's capture page: frame the result with that camera and press **Capture** (or the space bar) when you like it.
4. **The prints.** Kircher's cube is a room you carry with an artist inside it; Zahn's box is a camera with no film, 150 years early. The 8×10 in front of you is the same machine, and the projector is the same machine pointed the other way.

Notice

Every picture here is a chain: lens, glass, phone; or camera, screen, camera, screen; or stage, projector, wall, phone. Nothing in the chain captures a vision. Something arrives out of the lens, the sheet, the leak, the loop, and your hands, and none of them fully intended it. The vignette, the hot spot, the grain, the leak, the ghost are the marks the chain leaves. Keep the ones you like.